

Mathematical Statistics

Lecture 12: Understanding the Binomial Distribution

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November 18, 2025

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💡 Learning Objectives

Upon completing this lecture, you will be able to:

- Define the Binomial Distribution and its relationship to the Bernoulli distribution.
- Identify the four key conditions required for an experiment to be binomial.
- Understand and derive the Probability Mass Function (PMF) for the Binomial Distribution.
- Apply the binomial formula to solve real-world problems involving a fixed number of trials and two outcomes.
- Recognize the importance of the binomial distribution in fields like data science, quality control, and medicine.

1 Introduction to Binomial Distribution

The Binomial Distribution is a fundamental discrete probability distribution with vast applications in statistics and data analysis. It builds directly upon the concept of the **Bernoulli Distribution**.

- **Bernoulli Trial:** An experiment with exactly two possible outcomes: "success" and "failure". For instance, a single coin toss can result in Heads (success) or Tails (failure).

The Binomial Distribution extends this idea by modeling the number of successes in a **fixed number of repeated, independent Bernoulli trials**. It helps us answer questions like, "If I toss a coin 10 times, what's the chance of getting exactly 7 heads?" or "If 5% of products are defective, what is the probability that in a batch of 20, exactly 2 are defective?"

2 What is a Binomial Distribution?

A random variable X is said to follow a Binomial Distribution if it represents the total number of successes in an experiment that satisfies four key conditions.

2.1 Conditions for a Binomial Experiment

For an experiment to be modeled by a Binomial Distribution, it must satisfy all of the following criteria:

1. **Fixed Number of Trials (n):** The experiment consists of a predetermined number of trials. For example, tossing a coin 10 times ($n = 10$) or testing 50 products for defects ($n = 50$).
2. **Independent Trials:** The outcome of each trial must be independent of the others. The result of the first coin toss does not influence the result of the second.
3. **Two Possible Outcomes:** Each trial must result in one of only two outcomes, which we categorize as **success** or **failure**. For example, a medical test can be positive (success) or negative (failure).

4. **Constant Probability of Success (p):** The probability of success (p) must remain the same for every trial. The probability of failure is consequently $1 - p$ (often denoted as q).

3 The Probability Mass Function (PMF)

The **Probability Mass Function (PMF)** of a Binomial Distribution calculates the probability of getting *exactly* x successes in n trials.

The formula is given by:

$$P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}$$

Where:

- n : The total number of trials.
- x : The specific number of successes for which we want to find the probability.
- p : The probability of success in a single trial.
- $1 - p$: The probability of failure in a single trial.
- $\binom{n}{x}$: The binomial coefficient, which calculates the number of ways to choose x successes from n trials. It is calculated as $\frac{n!}{x!(n-x)!}$.

3.1 Deriving the PMF with an Example

Let's understand how this formula is constructed.

Problem: We toss a fair coin 6 times. What is the probability of getting exactly 2 Heads? (Here, Success = "Heads", Failure = "Tails").

- **Parameters:** $n = 6$ (six tosses), $x = 2$ (two heads), $p = 0.5$ (probability of heads), $1 - p = 0.5$ (probability of tails).

1. **Probability of one specific outcome:** Consider one possible sequence of 2 heads and 4 tails, for example: **H T T T T H**. Since the trials are independent, the probability of this specific sequence is:

$$P(H) \times P(T) \times P(T) \times P(T) \times P(T) \times P(H) = p \cdot (1 - p)^4 \cdot p = p^2(1 - p)^4$$

This corresponds to the $p^x(1 - p)^{n-x}$ part of the formula.

2. **Counting all possible outcomes:** The sequence above is just one way to get 2 heads. Other ways include **H H T T T T**, **T H T H T T**, etc. We need to count how many such combinations exist. This is the binomial coefficient $\binom{n}{x}$. In our case, this is $\binom{6}{2} = \frac{6!}{2!4!} = \frac{6 \times 5}{2 \times 1} = 15$ ways.

3. **Total Probability:** The total probability of getting exactly 2 heads is the probability of one specific sequence multiplied by the total number of ways it can occur.

$$P(X = 2) = \binom{6}{2} \cdot (0.5)^2(0.5)^4 = 15 \times (0.25) \times (0.0625) = \mathbf{0.234375}$$

3.2 The Full Distribution (for $n=6$, $p=0.5$)

By applying the PMF for each possible value of x (from 0 to 6), we get the full probability distribution.

Table 1: Probability Distribution for a Binomial Experiment with $n = 6, p = 0.5$.

Successes (x)	Probability $P(X = x)$	Calculation
0	0.015625	$\binom{6}{0}(0.5)^0(0.5)^6 = 1 \cdot 1 \cdot 0.015625$
1	0.093750	$\binom{6}{1}(0.5)^1(0.5)^5 = 6 \cdot 0.5 \cdot 0.03125$
2	0.234375	$\binom{6}{2}(0.5)^2(0.5)^4 = 15 \cdot 0.25 \cdot 0.0625$
3	0.312500	$\binom{6}{3}(0.5)^3(0.5)^3 = 20 \cdot 0.125 \cdot 0.125$
4	0.234375	$\binom{6}{4}(0.5)^4(0.5)^2 = 15 \cdot 0.0625 \cdot 0.25$
5	0.093750	$\binom{6}{5}(0.5)^5(0.5)^1 = 6 \cdot 0.03125 \cdot 0.5$
6	0.015625	$\binom{6}{6}(0.5)^6(0.5)^0 = 1 \cdot 0.015625 \cdot 1$
Total	1.0	Sum of all probabilities

4 Real-World Examples & Solved Problems

4.1 Example 1: Public Health (Coronavirus Testing)

Problem: Historical data shows that in 30,000 coronavirus tests, 18,000 were positive. If 10 new tests are conducted today, what is the probability that **at most 8** patients test positive?

Step-by-Step Solution:

1. **Verify Conditions & Identify Parameters:** The experiment fits the binomial model.

- p (Probability of success/positive test) = $18,000/30,000 = 0.6$.
- n (Number of trials) = 10.
- x (Number of successes) = "At most 8" means we need to find $P(X \leq 8)$.

2. **Formulate the Calculation:** Calculating $P(X = 0) + P(X = 1) + \dots + P(X = 8)$ is long. It is easier to use the **complement rule:** $P(A) = 1 - P(\text{not } A)$.

$$P(X \leq 8) = 1 - P(X > 8) = 1 - [P(X = 9) + P(X = 10)]$$

3. **Calculate Probabilities for the Complement:**

- $P(X = 9) = \binom{10}{9}(0.6)^9(0.4)^1 = 10 \cdot (0.010077) \cdot (0.4) \approx 0.0403$
- $P(X = 10) = \binom{10}{10}(0.6)^{10}(0.4)^0 = 1 \cdot (0.006046) \cdot 1 \approx 0.0060$

4. **Final Answer:**

$$P(X \leq 8) = 1 - (0.0403 + 0.0060) = 1 - 0.0463 = \mathbf{0.9537}$$

There is a 95.37% probability that at most 8 out of 10 patients will test positive.

4.2 Example 2: Sports Analytics (Target Practice)

Problem: The probability of a man hitting a target is $1/3$.

Part A: If he fires 5 times, what's the probability of hitting the target **at least twice**?

Part B: How many times must he fire so that the probability of hitting the target **at least once** is greater than 90%?

Solution - Part A: $n = 5, p = 1/3$. We need to find $P(X \geq 2)$.

$$P(X \geq 2) = 1 - P(X < 2) = 1 - [P(X = 0) + P(X = 1)]$$

- $P(X = 0) = \binom{5}{0} \left(\frac{1}{3}\right)^0 \left(\frac{2}{3}\right)^5 = 1 \cdot 1 \cdot \frac{32}{243} = \frac{32}{243}$

- $P(X = 1) = \binom{5}{1} \left(\frac{1}{3}\right)^1 \left(\frac{2}{3}\right)^4 = 5 \cdot \frac{1}{3} \cdot \frac{16}{81} = \frac{80}{243}$

$$P(X \geq 2) = 1 - \left(\frac{32}{243} + \frac{80}{243}\right) = 1 - \frac{112}{243} = \frac{131}{243} \approx \mathbf{0.5391}$$

Solution - Part B: We need to find the smallest integer n such that $P(X \geq 1) > 0.90$.

$$1 - P(X = 0) > 0.90 \implies 0.10 > P(X = 0)$$

$$P(X = 0) = \binom{n}{0} \left(\frac{1}{3}\right)^0 \left(\frac{2}{3}\right)^n = \left(\frac{2}{3}\right)^n$$

So we need to solve $\left(\frac{2}{3}\right)^n < 0.10$. Using trial and error (or logarithms):

- For $n = 5$, $\left(\frac{2}{3}\right)^5 \approx 0.132$ (not less than 0.10)
- For $n = 6$, $\left(\frac{2}{3}\right)^6 \approx 0.088$ (this IS less than 0.10)

He must fire at least **6 times**.

4.3 Example 3: Data Analysis (YouTube Video Engagement)

Problem: Over 30 days, a YouTube video received 300 views and 60 likes. What is the probability of getting **exactly 4 likes tomorrow**?

Step-by-Step Solution:

1. **Translate to Binomial Terms:** A trial is a user viewing the video, and a success is that user liking it.
2. **Calculate Parameters from Data:**
 - n (**Trials tomorrow**): Average views per day = $300/30 = 10$ views/day.
 - p (**Probability of a like**): Historical rate = $60/300 = 0.2$.
 - x (**Successes tomorrow**): We need exactly 4 likes, so $x = 4$.

3. **Apply the PMF:**

$$P(X = 4) = \binom{10}{4} (0.2)^4 (1 - 0.2)^{10-4} = \binom{10}{4} (0.2)^4 (0.8)^6$$

4. **Calculate the Result:**

$$\binom{10}{4} = \frac{10!}{4!6!} = 210$$

$$P(X = 4) = 210 \cdot (0.0016) \cdot (0.262144) \approx \mathbf{0.08808}$$

There is approximately an 8.81% chance of getting exactly 4 likes tomorrow.

Practical Applications in Data Science & ML

The Binomial Distribution is a cornerstone for many data science tasks:

- **A/B Testing:** Used to determine if a change (e.g., a new website button) leads to a statistically significant improvement in conversion rates. Each visitor is a trial; converting is a success.
- **Click-Through Rate (CTR) Modeling:** Modeling whether a user will click an online ad. Each ad impression is a Bernoulli trial.
- **Quality Control:** In manufacturing, used to model the number of defective items in a batch. For example, testing a sample of 100 light bulbs.
- **Churn Prediction:** Predicting whether a customer will cancel their subscription (churn). Each customer is a trial with two outcomes.

5 Summary of Key Ideas

- The Binomial Distribution models the **number of successes** in a fixed number of **independent Bernoulli trials**.
- **Four Conditions:** Fixed n , independent trials, two outcomes, and constant p .
- **PMF Formula:** $P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}$.
- Problems involving "**at least**" or "**at most**" are often best solved using the **complement rule**.
- It is a powerful tool for converting real-world scenarios into a probabilistic framework.

6 Practice Problems

1. A factory produces laptops, and 5% are defective. If a quality inspector randomly selects 10 laptops, what is the probability that exactly one is defective?
2. You are taking a 20-question multiple-choice quiz. Each question has four options, and you guess randomly on every question. What is the probability you get at least 3 questions right?
3. It is known that 80% of students who take a specific programming course pass. In a class of 15 students, what is the probability that more than 13 students pass?
4. A sales representative has a 30% chance of making a sale on any given call. How many calls must they make to be at least 95% certain of making at least one sale?
5. A basketball player makes 75% of her free throws. In a game, she attempts 12 free throws. What is the probability she makes exactly 9 of them?